SYDE252 - W05B - System Insights

Dr. Matt Borland

Insights Into System Behaviour

Systems Depend on their Characteristic Modes.

Response Time of a System is Determined by h(t).

Time Constant: Rise time = System Time Constant.

Time Constant and Filtering.

Resonance.

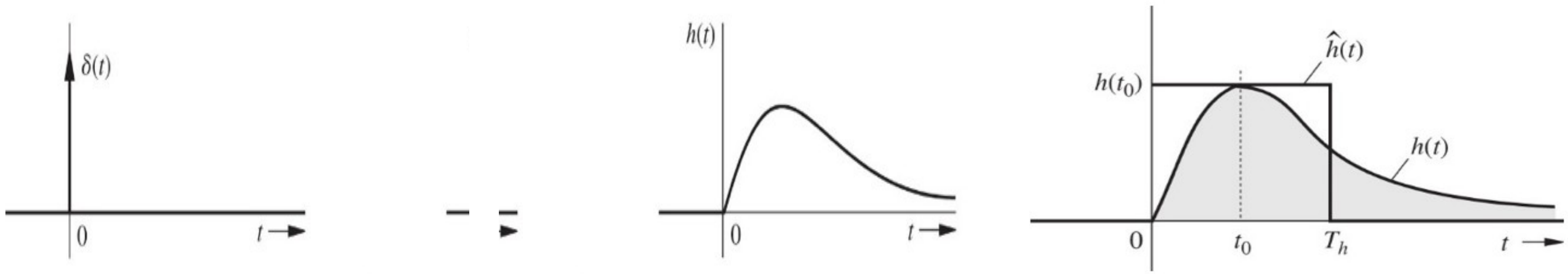
Systems Depend on their Characteristic Modes.

The ZIR of a system is a linear combination of the modes of the system!

https://youtu.be/wvJAqrUBF4w

$$y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \cdots + c_N e^{\lambda_N t}$$

Response Time of a System is Determined by h(t)



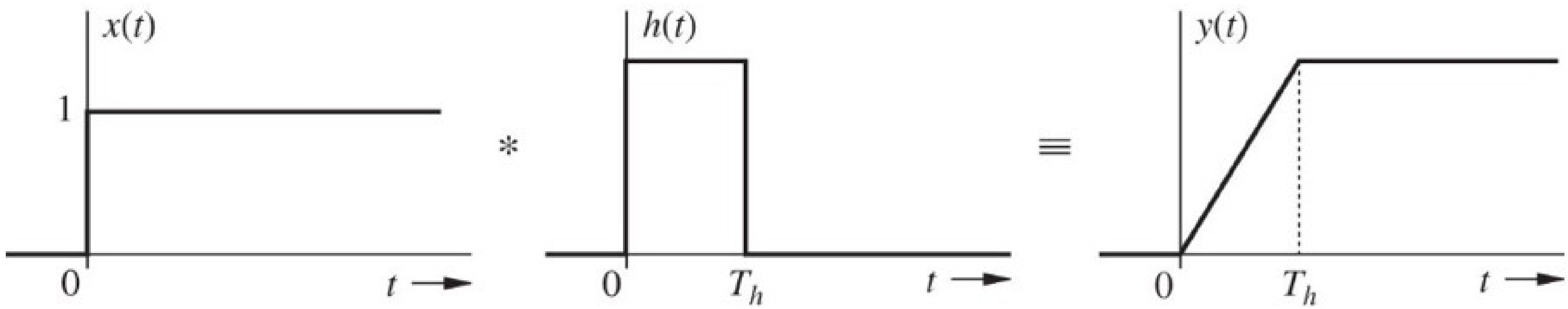
When an input is applied to a system, there is a certain amount of time, or lag, for that system to fully respond. This system response time, or lag, is sometimes called the system ***time constant***, T_h .

Smaller T_h means a faster responding system.

Longer T_h means a slower responding system.

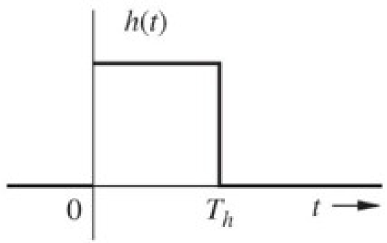
Time Constant: Rise time = System Time Constant.

The rise time of a system is defined (in our case) as the time required for the unit step response to rise from 10% to 90% of its steady-state value.



$$T_r = T_h$$

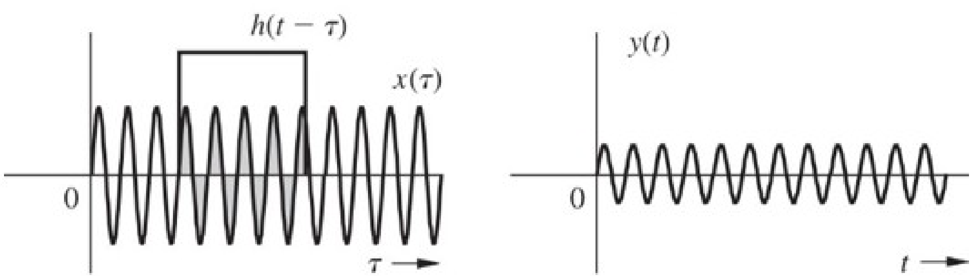
Time Constant and Filtering.



(a)

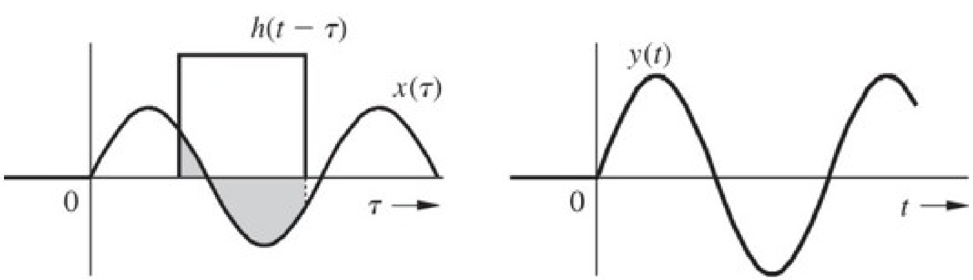
Smaller T_h means a faster responding system.

Longer T_h means a slower responding system.



(b)

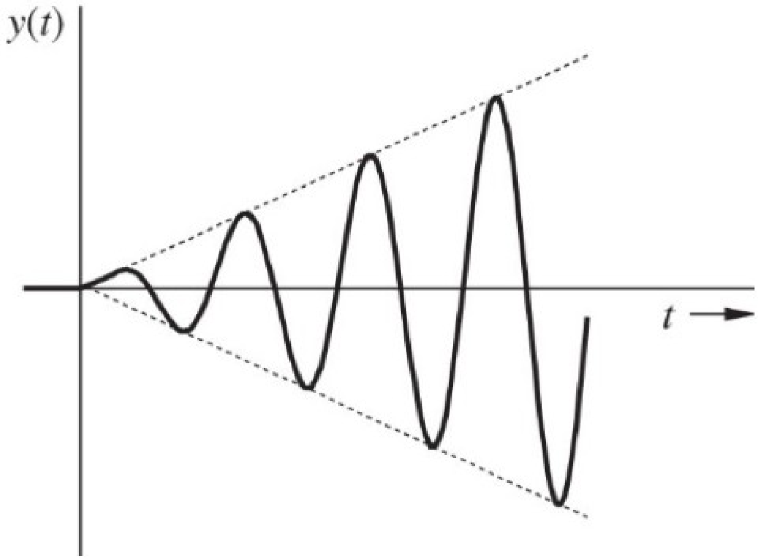
“Slow” systems can’t respond to quick (high frequency) changes to the input.



(c)

They can respond to slow (low frequency) changes, and functionally act as low-pass filters.

Resonance.



At frequencies corresponding to the modes of our system, the energy we input builds up, instead of dying out quickly. Resonance is cumulative, not instantaneous.

https://www.youtube.com/watch?v=k8Q1XAgHsz8

2.6-2 DETERMINE A FREQUENCY, ω, THAT WILL CAUSE $x(t) = \cos(\omega t)$ TO PRODUCE A STRONG RESPONSE WHEN APPLIED TO:

$$(D^2 + 2D + 13/4)\, y(t) = x(t) \qquad x(t) \to \boxed{S} \to y(t)$$

WE KNOW RESONANCE CAN OCCUR WHEN WE INPUT ENERGY AT FREQUENCIES CORRESPONDING TO THE MODES OF THE SYSTEM.

REMEMBER THE CHLADNI VIDEOS

SO LET'S FIND THE MODAL FREQUENCIES:

CHAR. EQN. $\lambda^2 + 2\lambda + 13/4 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-2 \pm \sqrt{4 - 13}}{2}$$

$$\lambda_{1,2} = -1 \pm j\frac{3}{2}$$

SO WE HAVE THE ROOTS CORRESPONDING TO THE MODES + NATURAL FREQ'S.

IF WE LOOK AT OUR INPUT

$$x(t) = \cos(\omega t) = 0.5\, e^{j\omega t} + 0.5\, e^{-j\omega t}$$

TO MATCH THE NATURAL FREQ OF THE SYSTEM WE USE THE IMAGINARY PORTION OF THE ROOT

$\cos(\omega t)$ WILL PRODUCE A STRONG RESULT WHEN $\omega = \pm 3/2$ RAD/S

I.E. WHEN WE EXCITE THE SYSTEM AT IT'S MODAL FREQUENCIES.